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Discontinuation of Medication Treatment for Opioid Use Disorder after a Successful Course: The Discontinuation Phase of the CTN-0100 (RDD) Trial

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Abstract

Introduction and Background: Buprenorphine, and extended-release naltrexone, are effective in decreasing opioid use, morbidity and mortality. The available evidence suggests that these medications should be used for long term treatment; however, patients often ask how long they need to be on medication, and whether it would be safe to discontinue. There are sparse data to guide us. The CTN-0100 trial will address this gap in our knowledge by studying participants who have decided to discontinue buprenorphine and extended-release naltrexone for OUD.

Research Design and Methods: The trial is a multicenter, randomized, non-blinded study. Participants are stable adult volunteers, on sublingual buprenorphine, extended-release buprenorphine, or extended-release naltrexone, expressing an interest in discontinuing medication. Participants on buprenorphine must be stable for at least 1 year and participants on extended-release naltrexone must be stable for at least 6 months. Participants are engaged in the study for up to 96 weeks, including a flexible taper period, and are then transitioned to follow-up within the

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Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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trial. All participants are randomly assigned to the study Medical Management (MM) or to MM plus *Connections* (CHESS health) digital smartphone application aimed at recovery and abstinence (MMD). Sublingual Buprenorphine participants are also randomized (2×2 design) to a taper using either sublingual or extended-release buprenorphine.

Discussion/Conclusion: It is hoped that this trial will provide a rich source of data on management of patients discontinuing medication for opioid use disorder (MOUD) to inform future research and practice. The trial will shed light on which strategies are most likely to lead to long-term success (absence of relapse), and what participant characteristics distinguish those who can safely discontinue MOUD from those who remain at risk of relapse should they discontinue.

Keywords

Treatment; Opioids; Buprenorphine; Naltrexone; Opioid Use Disorder; Automated psychosocial intervention

1.0 Introduction and Background

Opioid use disorder (OUD) is associated with substantial morbidity and mortality, with over 80,000 annual overdose deaths related to opioid use disorder in the U.S. in recent years^{1,2}. Long-term treatment with buprenorphine, or extended-release naltrexone is highly effective and the recommended approach to care³⁻⁶. Patients who remain on medication for at least a year appear to do better in terms of relapse, overdose and other recovery measures than those who do not⁷⁻⁹. In clinical practice, however, patients often ask when they can come off medications for opioid use disorder (MOUD) due to the stigma associated with these medications, insurance coverage or cost, inconvenience, side effects, or wanting to feel “cured” of OUD¹⁰. This creates a difficult clinical juncture for patient and clinician, since the risks of discontinuation of medication are relapse, overdose and death¹¹. There is limited evidence, or consensus, on best practices for the safe discontinuation of MOUD¹².

The Discontinuation Phase of the CTN-0100 Optimizing Retention, Duration, and Discontinuation Strategies for Opioid Use Disorder Pharmacotherapy (RDD) Study, funded by the NIH Helping to End Addiction Long-Term (HEAL) Initiative and the NIDA Clinical Trials Network (NIDA CTN), was designed to fill these evidence gaps, and better inform the shared decision-making between clinicians and patients on buprenorphine and extended-release naltrexone. Importantly, for safety reasons, participants are only recruited after they have expressed interest in stopping MOUD and, once enrolled, can take the taper of MOUD at their own pace and decide to abandon the discontinuation effort at any time.

The current study will test the CHESS Health *Connections* application to support long-term recovery among participants discontinuing buprenorphine or extended-release naltrexone (XR-naltrexone). No digital applications have been tested for support individuals who are stable and discontinuing treatment. *Connections* was chosen because it allows for use of digital peer support that may be particularly helpful for individuals already in recovery and is commercially available and easily scalable.

Connections is the commercial iteration of the technology based intervention *Addiction version of the Comprehensive Health Enhancement Support System*²², an evidence based²³ computer delivered intervention based on self-determination theory.¹⁷ *Connections* supports recovery through psychoeducational videos on discontinuing medication, motivational text messaging, connecting participants with their support network, and social networking with others in recovery. *Connections* also includes the evidence based^{24,25} digital cognitive behavioral therapy program CBT4CBT, which includes 7 modules teaching recovery skills.

The trial will also test the optimal approach to discontinuation of sublingual buprenorphine through randomization to either a standard slow taper of sublingual buprenorphine or a taper using XR-buprenorphine. Clinical-anecdotal reports^{18,19} suggest that many patients experience opioid withdrawal under the standard taper approach. Transition to XR-buprenorphine has potential advantages in terms of pharmacokinetics (blood levels decay gradually)^{20,21} and psychological expectation (no abrupt change in behavior).

Few studies have focused on predictors of long-term outcomes after discontinuation of MOUD. Thus, the best a clinician can advise to patients stable and abstinent on MOUD but who want to discontinue is that some will do well when medication is discontinued, but many will relapse and be at risk of overdose and death⁵. This trial will identify patient factors that are associated with maintained recovery after MOUD discontinuation.

2.0 Research Design, Objectives and Key Outcomes Measured

Research Design

This study is a multicenter, randomized, non-blinded study, consisting of two phases, a Retention Phase (for patients seeking MOUD as part of a new treatment episode), and a Discontinuation Phase (for patients who are stable on MOUD and have decided to discontinue). This paper focuses on the Discontinuation Phase protocol. For the Discontinuation Phase Design Schema, see Figure 1. The study was approved by a single Institutional Review Board (IRB), BRANY (Lake Success, NY).

1. Research Objectives, Aims, and Hypotheses

The primary objectives of the Discontinuation Phase are: 1) to test strategies to improve outcomes among patients who have achieved *stable remission* on MOUD and want to discontinue MOUD; and 2) to develop models to predict, based on patient characteristics and including duration of MOUD treatment, who is able to discontinue MOUD without relapse to opioid use.

The specific study aims include determining 1) the superior discontinuation method for patients on sublingual buprenorphine; 2) the impact of the *Connections* app; and 3) patient characteristics that may predict successful long-term outcomes after discontinuing MOUD (Table 1).

2.1 Key Outcomes Measured

The primary outcome, completed discontinuation without relapse (binary yes/no), is defined as discontinuing medication during the taper period, and no return to MOUD or relapse to

opioid use, either during the taper or during the 24 weeks after MOUD is discontinued. Relapse is defined as self-reported opioid use on >4 days in any consecutive 28-day period as captured on the Timeline Follow Back, or 2 or more consecutive opioid-positive drug tests or any self-reported injection drug use. Any opioid use was not chosen as the primary outcome as a single use event may not be clinically important, whereas this definition of relapse has been used in previous trials to capture clinically significant drug use.⁴

A key secondary outcome is the subcategorization of participants who do not meet the criteria for completed discontinuation without relapse into three categories: 1) did not complete discontinuation and did not relapse, 2) completed discontinuation followed by relapse, 3) did not complete discontinuation and relapsed (For full descriptions and definitions see Table 1).

3.0 Study Population and Treatments

1. Study Population, Inclusion/Exclusion Criteria

Candidates must have completed at least 1 year of buprenorphine treatment (Sublingual-buprenorphine or extended release (XR)-buprenorphine with the CAM2038 product (now marketed as BrixadiTM)) or 6 months of XR-naltrexone treatment. Individuals in this group are only eligible for the study if they actively express an interest in discontinuing their MOUD. Additionally, candidates must be stable in their recovery as evidenced by no opioid, cocaine, methamphetamine or non-prescribed benzodiazepine use, and no alcohol use rising to the level of DSM-5 criteria for alcohol use disorder, for at least the past 12 weeks. See Table 2 for the listing of all inclusion and exclusion criteria.

2. Study Recruitment

Candidates may be recruited for the Discontinuation Phase of this trial from: 1) the Retention Phase of this trial, 2) the standing caseloads of participating treatment programs, 3) an outside clinician, or 4) through self-referral. Patients are not encouraged to discontinue their MOUD for purposes of enrolling and are only told about the trial if they approach clinical or research staff stating they want to discontinue medication. Study consent is offered only after a shared decision-making^{6,18} conversation with the study medical clinician during which the candidate's values, preferences, reasons for wanting to discontinue, and overall treatment history are discussed, and the risks of discontinuation, including relapse, overdose, and death, are thoroughly reviewed.

3. Randomization

The randomization plan is managed by a central Data and Statistics Center, through an automated, randomized assignment, stratified by site and history of injection drug use (Yes/No). Participants maintained on SL-buprenorphine are randomized to both a pharmacologic condition (SL-buprenorphine taper or XR-buprenorphine taper) and a behavioral condition (Medical Management [MM] or Medical Management plus *Connections* [MMD]). Participants maintained on XR-buprenorphine (CAM2038, now BrixadiTM) or XR-naltrexone prior to entering the Discontinuation Phase are randomized only to a behavioral condition (MM or MMD).

4. Study Site Characteristics and Site Selection Process

This study involves 18 sites from across the NIDA CTN. The Discontinuation and Retention Phases of the study occur simultaneously within the same sites, using the same staff and research infrastructure. Prospective sites submitted responses to a site selection survey and participated in interviews with study investigators. Sites were selected based on geographic diversity, patient flow, and other resource considerations (Table 3).

5. Study Interventions

5.1 Study Schedule and Follow-up—Participants are engaged in the study for up to 96 weeks, including a flexible taper period (up to 48 weeks for those entering on SL-buprenorphine or XR-buprenorphine; up to 24 weeks for those entering on XR-naltrexone), and 24–48 weeks of follow-up. Participants are transitioned to follow-up at the point of 1) completing discontinuation within the taper period; 2) deciding not to taper and to remain on medication; 3) reaching the end of the taper period; or 4) meeting study relapse criteria.

5.2 Study Medical Management—Participants receive care at their treatment programs including study Medical Management (MM) plus usual care included other interventions and follow-up that the program typically offers. MM occurs every other week for the first 8 weeks of the trial and monthly thereafter and is delivered by clinician prescribers (medical doctors, nurse practitioners). To ensure fidelity to the protocol prescribers received extra training on study procedures, use study-provided standardized forms for visit documentation, and meet with the study lead team for weekly clinical meetings to discuss clinical and research questions that arise. At these meetings, the lead investigators emphasize that the primary goal of clinical care is to ensure patients do not relapse, not that they complete the taper. Study medications during the taper period are provided by the study at no cost to the participant. Participants may receive prescriptions for other adjunctive medications (e.g., clonidine) during the taper as clinically indicated, although these medications are not provided through the study. During post-taper follow-up, the participant may be referred for further clinical treatment at either the study site or another community site.

MM sessions focus on building patient-clinician rapport and partnership, education surrounding OUD and treatment, establishing and maintaining a plan for medication adherence during the taper, advice and encouragement to maintain abstinence, assessment for signs of relapse, monitoring medication side effects and dose adjustments, and support for ancillary treatment (see Table 4 for schedule of Medical Management sessions)^{4,19–21}.

5.3 Behavioral Intervention Group Assignments—All eligible participants in the Discontinuation Phase are randomly assigned, with equal probability, to the study Medical Management (MM) or to MM plus *Connections* (MMD). Participants randomized to MMD are provided with the *Connections* application until Week 24 of the follow-up period. Participants are not provided cell phones as part of the trial to match real world conditions. Participants are given instruction on how to use the app by research staff, and staff checks on and encourages app usage as part of the MM visits. There is no specific timeline for module completion.

5.4 Buprenorphine Tapering with SL-buprenorphine and Transition from SL-buprenorphine to XR-buprenorphine—

Participants who enter on SL-buprenorphine are therefore randomly assigned to either: 1) SL-buprenorphine taper, consisting of a gradual, flexible dose reduction schedule, individualized to the participant's response; or 2) transition to XR-buprenorphine (CAM2038, now Brixadi™) consisting of a starting dose and an option for subsequent tapering doses. We chose to study this formulation of XR-buprenorphine due to the range of injection doses available, which is helpful in facilitating the taper. The initial monthly dose of XR-buprenorphine is selected to be equivalent to the current SL-buprenorphine dose. Participants are permitted to use supplemental study supplied sublingual buprenorphine if they experience cravings or withdrawal on XR-buprenorphine alone.

5.5 Discontinuing XR-buprenorphine or XR-naltrexone—Participants who enter the Discontinuation Phase on XR-buprenorphine receive tapering doses based on joint decision-making. Individuals entering the study on XR-naltrexone may simply stop medication as no withdrawal symptoms are associated with naltrexone.

5.6 Protection with XR-naltrexone After Discontinuing Buprenorphine—

Participants completing a taper with buprenorphine are encouraged to take XR-naltrexone afterwards, as relapse prevention. The study supplies up to 6 doses of XR-naltrexone to participants who choose to initiate it in the follow-up period.

5.7 Provisions for Ongoing Care and Premature Withdrawal—In typical clinical practice, patients who discontinue MOUD frequently discontinue treatment altogether.²⁶ However, best clinical practice suggests that patients should continue to be followed to 1) identify relapse stressors, and 2) detect relapse early and re-engage the patients in MOUD. To minimize relapse, all participants in the Discontinuation Phase are followed regularly with MM visits with a study medical clinician after stopping MOUD. These MM visits are documented in the study data system as part of the trial delivered interventions. Participants who wish to return to MOUD are supplied with study SL-buprenorphine for up to 8 weeks while they are re-establishing community-based care.

5.8 Follow-Up—Participants are followed for the duration of the study unless they withdraw consent or the Investigator or Sponsor decides to terminate their enrollment for any reason (e.g., a risk to the safety of the participant or study staff arises). Participants are followed for 24 weeks after their taper period concludes. If possible, within the overall study timeline, participants are followed for an additional 24 weeks (48 weeks total) after the taper period. Tracking procedures are followed to locate and assess participants at study follow-up timepoints and to minimize missing data.

5.9 Participant Reimbursement—Study participants are provided with pharmacotherapy and MM or MMD at no cost, and they are reimbursed for their participation in accordance with IRB policies. Participants will receive reimbursement for remainder of the trial regardless of their taper status

4.0 Study Assessments

4.1 Screening and Baseline Assessments

Screening and baseline assessments collect demographic information, MOUD treatment and substance use history, and a variety of medical, social, and psychological characteristics (see Table 4).

4.2 Assessments During Taper and Follow-up

Assessments collect data on MOUD treatment, substance use, craving, withdrawal, psychosocial, medical, and psychiatric status, and study adherence measures (medication log, medical management log, missed visit and visit documentation, and protocol deviations). Throughout the study, safety events are assessed and recorded, including overdoses, medication injection site examinations/reactions, medication side effects, and pregnancies. For a full list of assessments, see Table 4. In addition, data on app usage is collected through data transfer from CHES health.

5.0 Data and patient safety monitoring

The study is overseen by a Data and Safety Monitoring Board which meets annually, at minimum, and reviews safety events and other study data to ensure participant safety and the integrity of data collection. Treatment clinicians at each site are responsible for monitoring participants for possible clinical deterioration or other problems, and for taking corrective action per established operating procedures.

6.0 Statistical Data Analysis

Rationale for Sample Size and Statistical Power

The sample size was originally planned at 500 participants entering the study on SL-buprenorphine. The sample size of 500 participants with SL-buprenorphine was selected, based on simulations, to have a power of 0.95 to detect a true difference of 15% between the SL-buprenorphine vs. XR-buprenorphine taper (Aim 1), and between the behavioral intervention conditions MM vs. MMD (Aim 2). The 15% difference was chosen based on a survey study²⁷ on the minimum difference addiction-treatment clinicians would consider clinically meaningful, and worth implementing a new practice. Recruitment challenges encountered to date suggest that a target sample size of $N = 310$ entering on SL-buprenorphine is more realistic, and should still yield acceptable, albeit lower, power of 0.80.

The likely availability of participants enrolling on XR-naltrexone or XR-buprenorphine was uncertain, however, because of the unique opportunity to collect long term follow-up data on these participants it was decided to enroll as many as possible, up to 250 in each group. These participants, in combination with those entering on SL-buprenorphine, would form an augmented secondary analysis of the behavioral treatment MM vs. MMD (described below).

Analysis of Demographic and Baseline Characteristics

Baseline demographic and clinical variables will be summarized by study site and treatment arm for all randomized participants. Comparisons of treatment groups with respect to baseline characteristics will be descriptive; however, if meaningful differences between treatments arms are suspected, statistical testing may be performed to assess significance. Subgroup analyses for sex, age, race and ethnicity will be performed as required by the National Institutes of Health. These subgroup analyses will utilize the same statistical model as for the primary outcome analysis (described below), but with the inclusion of an interaction term between treatment arm and the demographic subgroup. Contrasts will be used to test for statistically significant differences in the primary outcome hypothesis by subgroup.

Aims 1 and 2 Primary Outcome Analysis and Significance Testing

The primary outcome analysis for Aims 1 and 2 will be based on a logistic ANOVA with a binary stratifying covariate, a random site effect, and fixed row and/or column effects as appropriate.

The Discontinuation Phase cohorts (SL-buprenorphine, XR-buprenorphine, and XR-naltrexone arms) will be treated independently, with no effort to control family-wide error among them. However, the SL-buprenorphine Discontinuation arm is factorial in design, so the tests for row and column main effects within each of these trials will be controlled via the Bonferroni adjustment, demanding $p \leq 0.025$ for either main effect to be nominated as significant. Further, tests of main effects will be performed only if the prior interaction test is not significant at $\alpha = 0.05$. If the interaction is significant, exploration will be required to elucidate it. Dropouts will be treated as relapses, so that no adjustments will need to be made in the analyses described above to accommodate missing data.

Aims 1 and 2 Planned Secondary Analysis

The analysis of the behavioral component of the Discontinuation trial will be extended by appending data from participants who enter on XR-naltrexone or XR-buprenorphine. These participants will be added to the corresponding MM vs MMD participants from SL-buprenorphine, forming an augmented dataset for a secondary analysis of the main behavioral effect. This secondary analysis will employ the same statistical model from the primary outcome analysis and the main behavioral effect will be tested only if the interaction term (pharmacologic treatment x behavioral treatment) is non-significant. Simulations suggest that power to detect a main effect of the behavioral treatment may be enhanced by a few percentage points in this augmented analysis.

Aim 3 Predictor/ Moderator Exploration

Potential features predicting success of completed discontinuation without relapse may include not only the main treatment assignments of the trials, but also characteristics from the pre-specified set indicated in Table 5. Predictive regression models will be used to explore possible predictors. To obtain unbiased estimates of successful prediction rates and error, we will randomly divide the data into two sets before exploration: training and test. All exploration and model development will be performed on the training data,

with preliminary prediction and error rates constructed by cross-validation. The test data will be sequestered until all exploration is concluded and final models are chosen and parameterized. Unbiased estimates of prediction accuracy will be calculated by applying the final models to the test data. Based on expected final sample sizes, we plan a test set of 100 observations, which will guarantee a confidence interval half-width of no more than 0.1 for the unbiased estimate of prediction accuracy.

Other Secondary Outcomes Measured

Other Secondary Outcomes include relapse (time to event), withdrawal and craving (SOWS and OCS), medical and psychiatric data (depression, anxiety, stress, pain), as well as demographic data (homelessness, incarcerations), safety (overdoses, hospitalizations, study medication effects), and outcomes related to recovery capital, sexual risk taking, and negative consequences of opioid use. Analyses of these other secondary outcomes will follow a form similar to that of the primary outcome analysis, but with different linear models as appropriate: dichotomous outcomes will use logistic regression; time-to-event outcomes will use survival analysis with Cox models; and continuous or count outcomes will use normal, Poisson, negative binomial, or zero-inflated models as appropriate. Repeated measures will have time in the model in addition to treatment and baseline variates.

Safety Analysis

Adverse events (AEs), including serious adverse events (SAEs), will be summarized by body system and preferred term using the Medical Dictionary for Regulatory Activities. AEs, adverse drug events (ADEs), and events of special interest (EOSI) will be presented as: the number and proportion of participants experiencing at least one incidence of each event overall. Expected events (injection site reactions, withdrawal symptoms, overdoses, etc.) all of which are reported on study-specific forms, will be tabulated separately and not duplicate-reported as an AE. However, any study safety event that meets SAE criteria will be concurrently reported and eventually tabulated as an SAE.

7.0 Current Status of the Study and Challenges

The first participant was randomized into the study on June 15, 2021. As of October 2023, the study has enrolled 201 participants into the Discontinuation Phase (183 entering on SL-buprenorphine; 18 on XR-naltrexone or XR-buprenorphine) with study recruitment ongoing. Recruitment has been slow, not surprisingly, given the stringent eligibility criteria. Sites have reached outside their own caseloads, establishing communication with other treatment programs in their communities with large caseloads of MOUD patients.

The study encountered a challenge when the extended-release injectable buprenorphine formulation CAM2038 (now Brixadi™) became temporarily unavailable for a period of about 4 months, due to deficiencies at a third-party manufacturing facility. During this pause, for participants entering the study on sublingual buprenorphine, the protocol was temporarily modified to remove XR-buprenorphine tapering (i.e., tapering was on sublingual buprenorphine only), and randomized assignment was therefore simplified to the MM vs.

MMD randomization only. Once XR-buprenorphine was again available, the original study design and randomization scheme resumed. Not disrupting the randomization to MM vs. MMD during the pause preserved the study aims related to the behavioral treatment. In addition we continued to accrue participant data for examining predictors of long-term outcome and modelling of relapse risk. Although the pause in XR-buprenorphine availability may require some modification to the original analysis plan, a relatively small number (N=16) of participants were randomized under the modified scheme.

8.0 Discussion/Conclusion

The Discontinuation Phase of the CTN-0100/RDD Study is among the first prospective studies to systematically examine what happens when medication treatment for opioid use disorder (MOUD) is discontinued after a successful period of sustained stability. Observational studies have shown that the risk of relapse and overdose remain high even after long courses of MOUD.^{3–6} Thus, our key study questions are which patients can safely discontinue medication, and what pharmacological or behavioral strategies increase the likelihood of successful, safe discontinuation.

Human subjects protections issues considered during protocol development included the potential risks, including overdose and death, associated with MOUD discontinuation. Nonetheless, many patients on MOUD want to and often do discontinue medication. To mitigate these risks, eligibility is restricted to patients already maintained on MOUD and abstinent for a long period.

The study also offers regular medical monitoring as a safety feature. This offers the potential for rapid detection of relapse or relapse risk.

It was resolved that the study could not in any way encourage, let alone assign, patients to discontinue MOUD. Thus, eligibility criteria require a prospective participant to have already decided on their own to try to discontinue MOUD. Recruitment materials are crafted to make clinicians and prospective participants aware of the study while not encouraging discontinuation. The informed consent process emphasizes the risks of discontinuation and that the taper attempt as designed into the study procedures is self-directed and can be paused or halted by the participant at will.

Limitations include that the sample will be highly selected, representative of the most stable and adherent patients treated with MOUD. While this is an important first step in this line of research on discontinuation of MOUD, the findings will not directly apply to the many patients who are less adherent, decide to stop medication earlier in their course of treatment than would be admissible in this study, or exhibit less evidence of clinical stability (e.g., ongoing substance use). More research is needed across the spectrum of circumstances under which patients discontinue MOUD. The interventions were chosen based on limited prior data. Anecdotal reports have described withdrawal symptoms emerging after discontinuation of sublingual buprenorphine¹⁷. An extended-release formulation of buprenorphine (XR-buprenorphine; CAM2038), hypothesized to minimize withdrawal symptoms due to its slow rate of decrement in the blood, was chosen

to contrast with the SL-buprenorphine taper. Behavioral support in the form of a smartphone app (*Connections*) was chosen because of promising data from other clinical scenarios^{24,25}. More research is needed on the types of medical management and behavioral support that can improve safety and outcomes among patients discontinuing MOUD. It is expected this trial will provide a rich source of data on management of patients discontinuing MOUD to inform future research and practice.^{29,30}

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Highlights:

- Buprenorphine and XR-naltrexone are effective treatments and stopping them is risky.
- Little evidence exists on outcomes after stopping treatment in stable OUD patients.
- This trial tests an automated app based intervention to improve outcomes in such patients.
- It also will provide data to inform who is likely to do well after discontinuation.

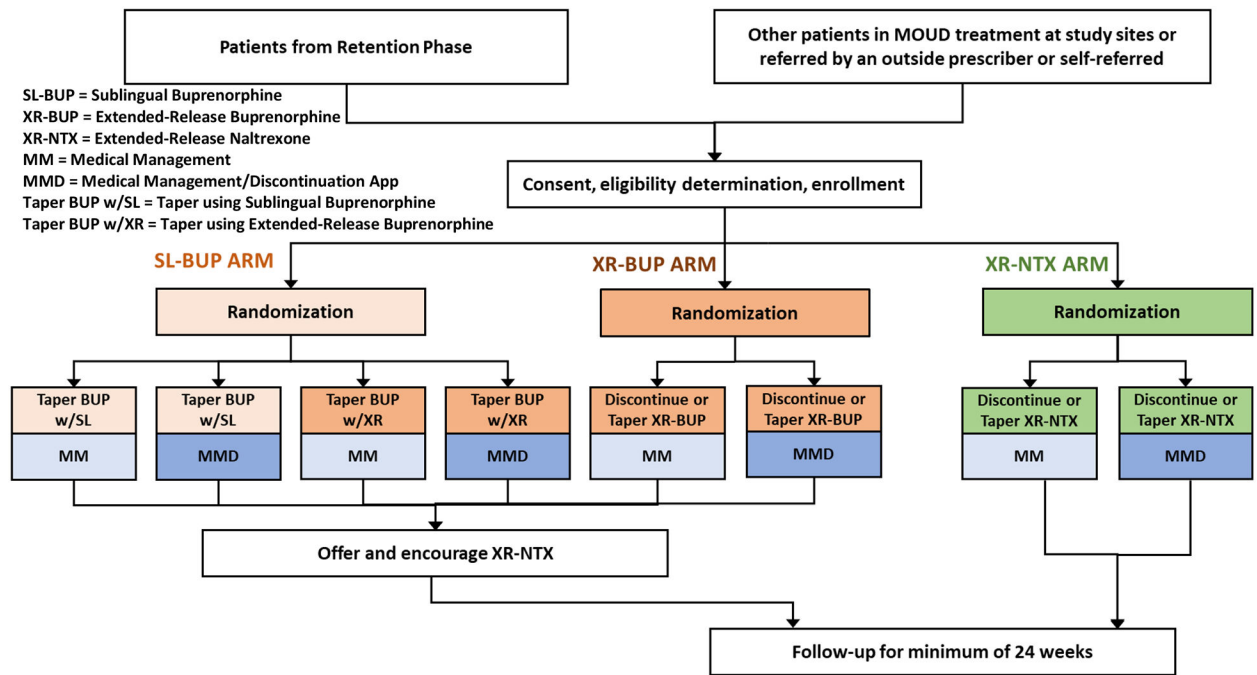


Figure 1:
Study Design Schema

Table 1.

Study Specific Aims and Outcomes

Specific Aim	Description
Superior Discontinuation Method for SL-buprenorphine	Among participants stable on SL-buprenorphine who plan to discontinue medication, to determine whether transition to XR-buprenorphine, which self-tapers, is superior to a standard SL-buprenorphine taper in improving the likelihood of Completed Discontinuation from buprenorphine without relapse to opioid use.
Impact of Technology-based Digital Therapeutic Intervention	1 Among participants stable on SL-buprenorphine who plan to discontinue medication, to determine whether access to a technology-based, digital therapeutic intervention, CHES Health <i>Connections</i>, added to usual Medical Management and counseling to support discontinuation (MMD) is superior to Medical Management (MM) alone in improving the likelihood of Completed Discontinuation from buprenorphine without relapse to opioid use. 2 Among participants stable on XR-buprenorphine or XR-naltrexone who plan to discontinue medication, to determine whether access to the technology-based intervention, <i>Connections</i>, added to usual Medical Management and counseling to support discontinuation (MMD) is superior to MM alone in improving the likelihood of Completed Discontinuation from buprenorphine or naltrexone without relapse to opioid use.
Patient Characteristics	Among stable participants discontinuing MOUD, to determine participant characteristics associated with the Completed Discontinuation without relapse outcome, and to develop a clinically useful predictive model that provides an estimate of likelihood that an individual will complete discontinuation without relapse. This predictive model would include participant characteristics at MOUD treatment entry (e.g., injection use vs. no injection use, other indicators of OUD severity); duration of MOUD; and current non-substance use status (e.g., recovery capital, distress tolerance, etc.). Hypotheses include that 1) longer duration of MOUD treatment prior to discontinuation ³¹ , and 2) lower severity of OUD prior to treatment entry (e.g., never IV use), and 3) higher degree of current recovery capital ³² , will be associated with greater likelihood of Completed Discontinuation without relapse to opioid use.
Outcomes	Description
Primary Outcome: <u>completed discontinuation without relapse (binary yes/no)</u>	Completed Discontinuation of MOUD during the taper period (48 weeks for buprenorphine, 24 weeks for XR-naltrexone) with no return to MOUD during the 24 weeks after MOUD was discontinued AND No relapse (relapse defined as self-reported opioid use on >4 days in any consecutive 28-day period as captured on the Timeline Follow back, or 2 or more consecutive opioid-positive drug tests or any self-reported injection drug use. ⁴)
Key Secondary Outcome: Subcategorize participants who do not meet the Primary Outcome criteria by treatment assignment	
Subcategory	Definition
Did not Complete Discontinuation and did not Relapse	MOUD is continued, or discontinued and then restarted within the next 24 weeks AND No relapse. This is considered a good or positive outcome.
Completed Discontinuation followed by Relapse	Completed Discontinuation of MOUD during the taper period (48 weeks for buprenorphine, 24 weeks for XR-naltrexone) with no return to MOUD during the 24 weeks after MOUD was discontinued FOLLOWED BY Relapse during the 24 weeks after MOUD is discontinued
Did Not Complete Discontinuation and Relapse	MOUD is continued, or discontinued and then restarted within the next 24 weeks AND Relapse to opioid use during the taper attempt or during the 24-week follow-up period

Table 2.**Participant Inclusion and Exclusion Criteria**

Inclusion Criteria	
1	18 years of age or older
2	Receiving buprenorphine for OUD for at least past year or XR-naltrexone pharmacotherapy for OUD for at least past 6 months prior to consent
3	Express desire to discontinue MOUD after shared decision-making discussion with the treating clinician
4	Meet stability criteria, i.e., have abstained from opioids (other than buprenorphine), cocaine, methamphetamine, and non-prescribed benzodiazepines for the past ≥12 weeks, and do not meet DSM-5 criteria for current (≥ 12 weeks) alcohol use disorder (participants with cannabis use will be eligible)
5	If currently taking buprenorphine, are willing to take either SL-buprenorphine or XR-buprenorphine if randomized to that condition
6	Willing to be randomized to either MM or to MMD
7	Able to provide written informed consent after discussion with their clinician regarding risks of discontinuation
8	Able to speak English sufficiently to understand study procedures
9	If female of childbearing potential, willing to practice an effective method of birth control while in the study and taking study medication (participants who become pregnant during the study will continue to be followed; treatment may be modified consistent with pregnancy).
Exclusion Criteria	
1	Serious medical or psychiatric disorder or concomitant medication that, in opinion of study medical clinician, would make study participation hazardous to participant or compromise study findings or would prevent participant from completing the study (e.g., uncompensated heart failure, cirrhosis or end-stage liver disease, active psychosis, or uncontrolled bipolar disorder requiring hospitalization).
2	Suicidal or homicidal ideation or behavior requiring different level of care (e.g., hospitalization)
3	For participants entering study taking buprenorphine, presence of pain requiring or likely requiring ongoing pain management with buprenorphine or other opioids
4	If female, currently pregnant or breastfeeding or planning on conception
5	Use of opioids (other than buprenorphine), cocaine, methamphetamine, or non-prescribed benzodiazepines in past 12 weeks
6	Meets DSM-5 criteria for any current alcohol use disorder
7	Currently in jail, prison or other overnight facility as required by court of law or have pending legal action that could prevent participation in study activities
8	Have used the <i>Connections</i> mHealth app in the 3 months prior to consent
9	Other major reasons that might prevent an individual from participating in the study (e.g., a planned move out of the area.

Table 3.

Site Selection Criteria

Selection Criterion	Description
MOUD Treatment	Treat large numbers of opioid use disorder patients with MOUD (buprenorphine and extended-release naltrexone (XR-naltrexone))
Facility Access	Have access to inpatient detoxification services and/or short-term residential treatment facilities to facilitate initiation of naltrexone if needed
Site Willingness	Are willing to participate in all elements of a randomized trial
Patients willing/ able to participate	Have enough current patients who are stable and want to discontinue MOUD to randomize at least 1–2 participants per month into the Discontinuation Phase of the study
Psychosocial Services	Provide basic psychosocial services either on-site, via telemedicine or by referral
Resources for Staff	Can provide adequate space and resources for at least 3 study-supported staff
Support for Uninsured	With support from the study budget, can enroll participants who do not have resources or insurance coverage to pay directly or through a third party for treatment
Required Staff on Site	Have on staff a licensed physician/medical clinician capable of completing all medical assessments, administering medical interventions, and providing medical treatment

Table 4.

Discontinuation Phase Schedule of Study Procedures and Assessments

DISCONTINUATION PHASE Schedule of Study Procedures and Assessments			S	B	Taper (up to 48 weeks)																Follow-Up Post-Taper (min of 24 weeks, up to 48 weeks)																EOT								
Timepoint	0		T1	T2	T4	T6	T8	T12	T16	T20	T24	T28	T32	T36	T40	T44	T48	varies	F1	F2	F4	F6	F8	F12	F16	F20	F24	F36	F48																
SCREENING AND ENROLLMENT																																													
Prisoner Status Assessment	X																																												
Informed Consent	X																																												
Medical Release	X																																												
Other Release of Information *	X																																												
Inclusion/Exclusion Checklist - Discontinuation Phase	X																																												
STUDY ASSESSMENTS																																													
Locator Form	X				X		X	X	X	X	X	X	X	X	X	X	X	X	X			X	X	X	X	X	X	X	X																
Demographics Form	X																																												
Study Demographics Form	X																X	X					X			X			X																
DSM-5 Checklist for AUD and CUD (current) ¹	X																																												
DSM-5 Checklist for SUDs (lifetime)	X																																												
Alcohol and Substance Use History	X																																												
Medical and Psychiatric History	+																																												
Physical Exam	+																																												
Vital signs	X																												X																
Pregnancy and Birth Control Assessment ²	X	*			X		X	X	X	X	X	X	X	X	X	X	*	*	*	*	*	*	*	*	*	*	*	*	*																
Liver Function Tests ³	*																*	*	*	*	*	*	*	*	*	*	*	*	*																
Genetics sample (if not drawn in the Retention Phase)	+																																												
Family Origin (if not completed in the Retention Phase)	+																																												
Drug Test	X		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X																
Timeline Followback ³³	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X																
Motivations for Discontinuation	X																																												
Treatment Effectiveness Assessment ³⁴		X															X	X									X																		

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DISCONTINUATION PHASE Schedule of Study Procedures and Assessments										S	B	Taper (up to 48 weeks)												EOT		Follow-Up Post-Taper (min of 24 weeks, up to 48 weeks)													
Timepoint											0	T1	T2	T4	T6	T8	T12	T16	T20	T24	T28	T32	T36	T40	T44	T48	varies	F1	F2	F4	F6	F8	F12	F16	F20	F24	F36	F48	
Post-taper XR-naltrexone administration (up to 6 injections)																												*	*	*	*	*	*	*	*	*	*	*	
Post-taper MOUD (up to 8 weeks) for transition back to community care																											*	*	*	*	*	*	*	*	*	*	*		
Medical Management											X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X							X	*	*	*	
mHealth <i>Connections</i> (MMD participants only)											X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
SAFETY ASSESSMENTS																																							
Safety Events (AE, ADE, SAE)											X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Mental Health Follow-Up Assessment											*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*
XR-buprenorphine Injection Site Exam & Reaction																																							
Exam conducted at the visit following each injection, and as needed.																																							
XR-naltrexone Injection Site Exam																																							
Exam conducted at the visit following each injection, and as needed.																																							
Study Medication Side Effects		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	*	*	*	*	*	*	*	*	*	*			
Overdose Questionnaire ⁴		X	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*			
SOWS		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X				
Confirmed Pregnancy form set		*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*			
COMPLIANCE MEASURES																																							
Participant Medication Log		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	*	*	*	*	*	*	*	*	*	*			
Medical Management Log	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X				
Missed Visit and Visit Documentation	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X				
Phone Documentation (MMD participants only)		X																								X								X					
Protocol Deviation Log	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*			
END OF TAPER / END OF STUDY																																							
Treatment Satisfaction Form																																							
End of Taper																																							
Study Completion Form																																							

S=Screening; B=Baseline; EOT=End of Taper

* As needed

⁴ Expected for participants entering from the clinic population (i.e., who did not participate in Retention Phase)

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[#]Complete once at final follow-up

¹ All participants (including those entering from Retention phase) will complete the DSM-5 for alcohol and cannabis use disorder only

² Repeated for as long as the participant receives study medication (every 4 weeks for XR-buprenorphine; at induction and prior to every monthly injection for XR-buprenorphine and XR-naltrexone). If participant initiates XR-naltrexone during the Follow-Up Post-Taper period, perform a pregnancy test prior to each injection.

3 Collect once for participants choosing to start XR-naltrexone

4 Assessed as scheduled and at any point in the study that an OD occurs

Table 5.
Pre-Specified Predictors (Features) of completed discontinuation without relapse

Predictor-type	Factor
Pre-treatment characteristics	Sociodemographic (gender; race; ethnicity; homelessness and living arrangements; criminal-legal involvement) Substance-use history - Opioid use - Intravenous versus non-intravenous - Prescription versus heroin/fentanyl - Other substance use Psychiatric comorbidities History of suicidal behavior
Length and nature of current substance use recovery status	How long taking MOUD How long opioid abstinent How long abstinent from other drugs Alcohol use status Cannabis use status
Other aspects of current status	Recovery capital * Stress Distress tolerance Chronic Pain Motivations for and expectations about discontinuation Grit ** Withdrawal symptoms Use of the digital therapeutic mobile application Psychosocial counseling 12-step programs and Medical Management

* a strength-based assessment that examines the factors associated with the recovery process that may be protective in the maintenance of recovery. As summarized by Vilsaint et al. (2017)³⁹, “recovery capital represents the quantity and quality of internal and external resources that can be brought to bear to initiate and sustain recovery from substance use disorders.”

** a predictor of dropout from challenging tasks